STAT 429: Introduction to Time Series
Week 10: Stationarity and AR(1) Models

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Adapted from Aditya Guntuboyina's time series lecture notes.

1 Time Series Models and Stationarity

Time series models describe observed time series data $y_1, \ldots, y_n$. These models describe the distribution of $y_1, \ldots, y_n$ in terms of various parameters. Even though, the observed data only corresponds to times $t = 1, \ldots, n$, it makes sense for the model to describe the distribution of y_t for all t past and present i.e., $t = \ldots, -3, -2, -1, 0, 1, 2, 3, \ldots$. This is because one may be interested in predicting the values of y_t for unobserved times.

The models discussed here have error terms ϵ_t that we assume are i.i.d Gaussian $N(0, \sigma^2)$. For these linear Gaussian models, the time series $\{y_t\}$ is jointly Gaussian. Gaussianity ensures that the behavior of the time series model is characterized by means (expectations) and covariances.

Some time series models satisfy the property of **stationarity** which is defined as follows:

Definition 1.1 (Stationarity)**.** *A time series model for $y_t, t = \ldots, -3, -2, -1, 0, 1, 2, 3, \ldots$ is said to be stationary if all the following conditions hold:*

1. *The mean of y_t (denoted by $\mathbb{E}y_t$) is the same for all times t*
2. *The variance of y_t (denoted by var(y_t)) is the same for all times t*
3. *The covariance between y_{t_1} and y_{t_2} only depends on the distance $|t_1 - t_2|$ between t_1 and t_2.*

Stationarity implies, for example, that the mean of y_{-2000} should be the same as y_{9999}. Also the covariance between y_{-2000} and y_{-2100} should be the same as the covariance between y_{9899} and y_{9999} etc.

For a stationary time series model $\{y_t\}$, the covariance between y_t and y_{t+h} will only depend on $|h|$. We denote:

$$\gamma(h) = \text{cov}(y_t, y_{t+h}) \qquad \text{for } h = \ldots, -2, -1, 0, 1, 2, \ldots.$$

$\gamma(h)$ is called the AutoCovariance Function (ACVF) of the stationary time series model $\{y_t\}$. Observe that

$$\gamma(0) = \text{cov}(y_t, y_t) = \text{var}(y_t) \;\&\; \gamma(-h) = \text{cov}(y_t, y_{t-h}) = \text{cov}(y_{t-h}, y_t) = \text{cov}(y_{t-h}, y_{t-h+h}) = \gamma(h)$$

So $\gamma(h)$ is a symmetric function of h, and we only need to evaluate it at nonnegative h.

The AutoCorrelation Function (ACF) of a stationary time series model $\{y_t\}$ is defined as:

$$\rho(h) = \text{correlation between } y_t \text{ and } y_{t+h} = \frac{\text{cov}(y_t, y_{t+h})}{\sqrt{\text{var}(y_t)\text{var}(y_{t+h})}} = \frac{\gamma(h)}{\sqrt{\gamma(0)}\sqrt{\gamma(0)}} = \frac{\gamma(h)}{\gamma(0)}.$$

Note that $\rho(0) = 1$ and $\rho(h) = \rho(-h)$.

It is important to remember the following points:

1. Stationarity refers to a time series model (not to actual data)
2. Not all time series models are stationary. In fact, there are many time series models that are not stationary.
3. ACVF and ACF are only defined for stationary time series models.

Let us now look at some examples of time series models, starting with the simplest.

Example 1.2 (Gaussian White Noise)**.** *The simplest time series model is $y_t = \epsilon_t$ where $\epsilon_t \overset{i.i.d}{\sim} N(0, \sigma^2)$. This is known as the Gaussian white noise model. It is easy to check that $\mathbb{E}y_t = 0$ and $cov(y_t, y_{t+h}) = \sigma^2 I\{h = 0\}$. So the conditions of stationarity are satisfied, and the Gaussian white noise is a stationary model. Its ACF is $\rho(h) = I\{h = 0\}$.*

Example 1.3 (Constant mean plus White Noise)**.** *The next simplest time series model is $y_t = \mu + \epsilon_t$ where, again, $\epsilon_t \overset{i.i.d}{\sim} N(0, \sigma^2)$. This is also a stationary time model because $\mathbb{E}y_t = \mu$ and $cov(y_t, y_{t+h}) = \sigma^2 I\{h = 0\}$. Its ACF is $\rho(h) = I\{h = 0\}$.*

The next two examples are for non-stationary time series models.

Example 1.4. *$y_t = \beta_0 + \beta_1 t + \epsilon_t$. Here the mean of y_t is:*

$$\mathbb{E}y_t = \beta_0 + \beta_1 t$$

and covariances are:

$$var(y_t) = \sigma^2 \quad \text{and} \quad cov(y_{t_1}, y_{t_2}) = 0 \ \text{ for } t_1 \neq t_2.$$

So the mean changes with t, variance is constant and there is no correlation between different time points. Because the mean changes with t, this is a non-stationary model.

Example 1.5. *$y_t = \beta_0 + \beta_1 \cos(2\pi f t) + \beta_2 \sin(2\pi f t) + \epsilon_t$ The means are given by:*

$$\mathbb{E}y_t = \beta_0 + \beta_1 \cos(2\pi f t) + \beta_2 \sin(2\pi f t)$$

and covariances are:

$$var(y_t) = \sigma^2 \quad \text{and} \quad cov(y_{t_1}, y_{t_2}) = 0 \ \text{ for } t_1 \neq t_2.$$

Again the mean changes with t, variance is constant and there is no correlation between different time points. Because the mean changes with t, this is non-stationary.

The next example is the spectrum model.

Example 1.6. *Consider the Spectrum model:*

$$y_t = \beta_0 + \sum_{j=1}^{m} \left(\beta_{1j} \cos \frac{2\pi jt}{n} + \beta_{2j} \sin \frac{2\pi jt}{n} \right) \tag{1}$$

with $\beta_{11}, \beta_{21}, \beta_{12}, \beta_{22}, \ldots, \beta_{1m}, \beta_{2m}$ *all independent with*

$$\beta_{1j}, \beta_{2j} \overset{i.i.d}{\sim} N(0, \tau_j^2).$$

Here m *is the largest positive integer that is strictly smaller than* $n/2$.

For this model, $\mathbb{E}y_t = \beta_0$ *so that the mean is constant over time (unlike the previous two models). The covariance is given by*

$$\begin{aligned}
&cov(y_{t_1}, y_{t_2}) \\
&= cov\left(\beta_0 + \sum_{j=1}^{m} \left(\beta_{1j} \cos \frac{2\pi jt_1}{n} + \beta_{2j} \sin \frac{2\pi jt_1}{n} \right), \beta_0 + \sum_{j=1}^{m} \left(\beta_{1j} \cos \frac{2\pi jt_2}{n} + \beta_{2j} \sin \frac{2\pi jt_2}{n} \right) \right) \\
&= \sum_{j=1}^{m} \left\{ cov\left(\beta_{1j} \cos \frac{2\pi jt_1}{n}, \beta_{1j} \cos \frac{2\pi jt_2}{n} \right) + cov\left(\beta_{2j} \sin \frac{2\pi jt_1}{n}, \beta_{2j} \sin \frac{2\pi jt_2}{n} \right) \right\} \\
&= \sum_{j=1}^{m} \left\{ \tau_j^2 \cos \frac{2\pi jt_1}{n} \cos \frac{2\pi jt_2}{n} + \tau_j^2 \sin \frac{2\pi jt_1}{n} \sin \frac{2\pi jt_2}{n} \right\} \\
&= \sum_{j=1}^{m} \tau_j^2 \cos \frac{2\pi j(t_1 - t_2)}{n} = \sum_{j=1}^{m} \tau_j^2 \cos \frac{2\pi j|t_1 - t_2|}{n}
\end{aligned}$$

This model incorporates dependence between y_t *at different time points* t *(unlike the previous two models). Further, the covariance between* y_{t_1} *and* y_{t_2} *only depends on the distance* $|t_1 - t_2|$ *between the two time points.*

A slightly inelegant thing about this model is that the specification (1) *depends on the sample size* n. *One can make the frequencies to take values in the continuum* $(0, 1/2)$ *and replace the sum* $\sum_{j=1}^{m}$ *by an integral. This can be done for example as in Shumway and Stoffer [1, Theorem C.2].*

Our next example is AR models. Are AR models stationary? The answer is a bit complicated. We focus on AR(1).

2 Stationarity of AR(1)

The AR(1) equation is

$$y_t = \phi_0 + \phi_1 y_{t-1} + \epsilon_t \tag{2}$$

One issue is that this equation does not fully specify y_t and there can be multiple processes $\{y_t\}$ that satisfy (2):

1. Suppose $\phi_1 \neq 0$ and $y_0 = 10$. Define $y_1, y_2, y_3, \ldots$ recursively via (2). Also define $y_{-1}, y_{-2}, y_{-3}, \ldots$ recursively via the following equation for $t = 0, -1, -2, \ldots$:

$$y_{t-1} = -\frac{\phi_0}{\phi_1} + \frac{y_t}{\phi_1} - \frac{\epsilon_t}{\phi_1} \tag{3}$$

Note that (3) is just a restatement of (2) obtained by rearranging (2) with y_{t-1} on the left hand side. The resulting time series model will then clearly satisfy (2). However it will not be stationary because:

$$\text{var}(y_0) = 0 \quad \text{and} \quad \text{var}(y_1) = \text{var}(\phi_0 + \phi_1 y_0 + \epsilon_1) = \text{var}(\phi_0 + \epsilon_1) = \text{var}(\epsilon_1) = \sigma^2.$$

2. Suppose $|\phi_1| < 1$ and define

$$y_t = \frac{\phi_0}{1-\phi_1} + \sum_{j=0}^{\infty} \phi_1^j \epsilon_{t-j}. \tag{4}$$

The summation in the right hand side above is an infinite summation, and hence we need to address convergence issues. Because $|\phi_1| < 1$, the terms ϕ_1^j rapidly decay to 0 as j increases. This ensures that $\sum_{j=0}^{\infty} \phi_1^j \epsilon_{t-j}$ is well-defined.

It is easy to check that (4) satisfies (2) because:

$$\begin{aligned}
y_t &= \frac{\phi_0}{1-\phi_1} + \sum_{j=0}^{\infty} \phi_1^j \epsilon_{t-j} \\
&= \frac{\phi_0}{1-\phi_1} + \epsilon_t + \phi_1 \epsilon_{t-1} + \phi_1^2 \epsilon_{t-2} + \phi_1^3 \epsilon_{t-3} + \dots \\
&= \frac{\phi_0}{1-\phi_1} + \epsilon_t + \phi_1 \left(\epsilon_{t-1} + \phi_1 \epsilon_{t-2} + \phi_1^2 \epsilon_{t-3} + \dots \right) \\
&= \frac{\phi_0}{1-\phi_1} + \epsilon_t + \phi_1 \left(y_{t-1} - \frac{\phi_0}{1-\phi_1} \right) = \phi_0 + \phi_1 y_{t-1} + \epsilon_t.
\end{aligned}$$

It is also true that (4) is a stationary model. This is because

$$\mathbb{E} y_t = \frac{\phi_0}{1-\phi_1} \qquad \text{for all } t$$

and, for $h \geq 0$,

$$\begin{aligned}
\text{cov}(y_t, y_{t+h}) &= \text{cov}\left(\frac{\phi_0}{1-\phi_1} + \sum_{j=0}^{\infty} \phi_1^j \epsilon_{t-j}, \frac{\phi_0}{1-\phi_1} + \sum_{k=0}^{\infty} \phi_1^k \epsilon_{t+h-k} \right) \\
&= \text{cov}\left(\sum_{j=0}^{\infty} \phi_1^j \epsilon_{t-j}, \sum_{k=0}^{\infty} \phi_1^k \epsilon_{t+h-k} \right) = \sum_{j=0}^{\infty} \sum_{k=0}^{\infty} \phi_1^{j+k} \text{cov}\left(\epsilon_{t-j}, \epsilon_{t+h-k} \right)
\end{aligned}$$

Because $\text{cov}\left(\epsilon_{t-j}, \epsilon_{t+h-k} \right)$ is non-zero (equal to σ^2) only when $t-j = t+h-k$ i.e., $k = j+h$, we get

$$\text{cov}(y_t, y_{t+h}) = \sigma^2 \sum_{j=0}^{\infty} \phi_1^{2j+h} = \sigma^2 \frac{\phi_1^h}{1-\phi_1^2}.$$

This clearly shows that $\{y_t\}$ is stationary with ACVF and ACF given by:

$$\gamma(h) = \sigma^2 \frac{\phi_1^{|h|}}{1-\phi_1^2} \text{ and } \rho(h) = \frac{\gamma(h)}{\gamma(0)} = \phi_1^{|h|}.$$

We have thus proved that (4) is a stationary time series model that satisfies the AR(1) equation (2) when $|\phi_1| < 1$. In fact, it turns out that (4) is the only stationary solution of (2) when $|\phi_1| < 1$ (I am skipping proof of this). Thus (4) is the unique stationary AR(1) model when $|\phi_1| < 1$. The model (4) when $|\phi_1| < 1$ is known as the **causal stationary** AR(1) model. Causal refers to the fact that y_t is fully determined by present and past values of $\{\epsilon_t\}$.

3. Suppose $|\phi_1| > 1$ and define

$$y_t = \frac{\phi_0}{1-\phi_1} - \sum_{j=1}^{\infty} \frac{\epsilon_{t+j}}{\phi_1^j}. \tag{5}$$

Note that y_t is well-defined because the infinite sum above has the coefficients ϕ_1^{-j} which decay rapidly as $|\phi_1| > 1$. It is easy to check that (5) also satisfies the AR(1) equation (2) and is stationary. In fact, it is the unique stationary AR(1) for $|\phi_1| > 1$. The model (5) is called the **non-causal, stationary** AR(1). It is non-causal because y_t depends on the future values of $\epsilon_{t+1}, \epsilon_{t+2}, \ldots$.

For the model (5), it is certainly not true that ϵ_t is independent of $y_{t-1}, y_{t-2}, y_{t-3}, \ldots$. The usual conditional likelihood for AR(1) assumes that ϵ_t is independent of past observations. Thus, if we use this likelihood to estimate the parameters ϕ_0, ϕ_1, σ of (5), we will obtain an incorrect estimate of ϕ_1 (for more details, see Example 3.3 and 3.4 in the Shumway-Stoffer book 4th edition).

To summarize the above discussion, there exist many non-stationary AR(1) time series models. When $|\phi_1| < 1$, there exists a unique stationary AR(1) model that is given by the formula (4), this is called the causal, stationary AR(1) model. When $|\phi_1| > 1$, there also exists a unique stationary AR(1) model that is given by the formula (5), this is called the non-causal, stationary AR(1) model.

When $|\phi_1| = 1$ (i.e., $\phi_1 = 1$ or $\phi_1 = -1$), neither of the two formulae (4) and (5) are meaningful (i.e., the infinite series do not converge). Here it turns out that there is no stationary AR(1) model. To see this, consider the case $\phi_1 = 1$ (the case $\phi_1 = -1$ is similar) where

$$y_t = \phi_0 + y_{t-1} + \epsilon_t$$

This implies that for every $t \geq 1$

$$y_t - y_0 = t\phi_0 + \epsilon_1 + \cdots + \epsilon_t$$

When $\phi_0 \neq 0$, clearly y_t and y_0 have different means (because $\mathbb{E}y_t = \mathbb{E}y_0 + t\phi_0$) so there is no stationarity. But even if $\phi_0 = 0$, we have

$$\text{var}(y_t - y_0) = \text{var}(\epsilon_1 + \cdots + \epsilon_t) = t\sigma^2$$

which approaches ∞ as $t \uparrow \infty$. But if $\{y_t\}$ were stationary, we would have

$$\text{var}(y_t - y_0) \leq 2\text{var}(y_t) + 2\text{var}(y_0) \leq \text{constant}.$$

Thus there are two kinds of AR(1): stationary and non-stationary. Stationarity is only possible when $|\phi_1| \neq 1$. There are also two kinds of stationary AR(1) models. When $|\phi_1| < 1$, the stationary AR(1) model has the formula (4); this is the causal kind of stationarity. When $|\phi_1| > 1$, the stationary AR(1) model has the formula (5); this is the non-causal kind of stationarity.

3 Deriving the Stationary AR(1) Solutions and Practical Implications

The AR(1) difference equation is $y_t = \phi_0 + \phi_1 y_{t-1} + \epsilon_t$. This is an implicit equation because y appears in both sides of the equation. To be able to calculate mean, variance and covariance etc., we need a more explicit formula for y_t.

One way to get such an explicit formula is to apply the difference equation recursively on the right hand side as:

$$\begin{aligned} y_t &= \phi_0 + \phi_1 y_{t-1} + \epsilon_t \\ &= \phi_0 + \phi_1 \left(\phi_0 + \phi_1 y_{t-2} + \epsilon_{t-1}\right) + \epsilon_t \\ &= \phi_0(1+\phi_1) + \phi_1^2 y_{t-2} + \epsilon_t + \phi_1 \epsilon_{t-1} \\ &= \phi_0(1+\phi_1) + \phi_1^2 \left(\phi_0 + \phi_1 y_{t-3} + \epsilon_{t-2}\right) + \epsilon_t + \phi_1 \epsilon_{t-1} \\ &= \phi_0(1+\phi_1+\phi_1^2) + \phi_1^3 y_{t-3} + \epsilon_t + \phi_1 \epsilon_{t-1} + \phi_1^2 \epsilon_{t-2}. \end{aligned}$$

We continue in this way writing $y_{t-3} = \phi_0 + \phi_1 y_{t-4} + \epsilon_{t-3}$ and then $y_{t-4} = \phi_0 + \phi_1 y_{t-5} + \epsilon_{t-4}$ and so on. This leads to

$$y_t = \phi_0 \sum_{j=0}^{M} \phi_1^j + \phi_1^{M+1} y_{t-M-1} + \sum_{j=0}^{M} \phi_1^j \epsilon_{t-j}. \tag{6}$$

This formula is true for every value of $M \geq 0$ and every ϕ_0, ϕ_1. The right hand side of (6) still depends on a y-value (specifically y_{t-M-1}).

Suppose now that $|\phi_1| < 1$ and that $\{y_t\}$ is stationary. Then ϕ_1^{M+1} decays rapidly to zero. In this case, we can let $M \to \infty$ in (6) and use

$$\begin{aligned} &\phi_0 \sum_{j=0}^{M} \phi_1^j \to \phi_0 \sum_{j=0}^{\infty} \phi_1^j = \frac{\phi_0}{1-\phi_1} \qquad \text{as } M \to \infty \\ &\phi_1^{M+1} y_{t-M-1} \to 0 \qquad \text{as } M \to \infty \\ &\sum_{j=0}^{M} \phi_1^j \epsilon_{t-j} \to \sum_{j=0}^{\infty} \phi_1^j \epsilon_{t-j} \qquad \text{as } M \to \infty \end{aligned}$$

Thus taking $M \to \infty$ in (6) leads to

$$y_t = \frac{\phi_0}{1-\phi_1} + \sum_{j=0}^{\infty} \phi_1^j \epsilon_{t-j}. \tag{7}$$

This expression is well-defined when $|\phi_1| < 1$. It is easy to check that y_t defined as (7) satisfies the AR(1) equation $y_t = \phi_0 + \phi_1 y_{t-1} + \epsilon_t$. As shown above, (7) is a stationary time series model with:

$$\mathbb{E} y_t = \frac{\phi_0}{1-\phi_1} \quad \text{and} \quad \operatorname{cov}(y_t, y_{t+h}) = \sigma^2 \frac{\phi_1^{|h|}}{1-\phi_1^2}.$$

In the formula (7), it is clear that y_t is determined by $\epsilon_t, \epsilon_{t-1}, \ldots$ i.e., by the present and past values of ϵ_t. For this reason, (7) is called **Causal, Stationary AR(1)** (causal here roughly means that y_t is fully determined by present and past values of $\{\epsilon_t\}$). One consequence of this kind of causality is that ϵ_t is independent of the past y-values $y_{t-1}, y_{t-2}, y_{t-3} \ldots$. (this is because these past y-values only depend on $\epsilon_{t-1}, \epsilon_{t-2}, \ldots$ which are all independent of ϵ_t).

When $|\phi_1| \geq 1$, the terms in the right hand side of (6) do not converge when $M \to \infty$. In this case, it is not possible to get a stationary solution y_t of the AR equation which depends only on present and past ϵ-values $\epsilon_t, \epsilon_{t-1}, \ldots$. In other words, there is no causal stationary solution to the AR(1) equation when $|\phi_1| \geq 1$.

When $|\phi_1| > 1$ (note the strict inequality), there is a non-causal stationary solution for the AR(1) equation. This is obtained by rewriting the AR(1) equation as:

$$y_t = -\frac{\phi_0}{\phi_1} + \frac{1}{\phi_1} y_{t+1} - \frac{\epsilon_{t+1}}{\phi_1}.$$

and then recursively plugging in $y_{t+1}, y_{t+2}, \ldots$ as:

$$\begin{aligned}
y_t &= -\frac{\phi_0}{\phi_1} + \frac{1}{\phi_1}\left(-\frac{\phi_0}{\phi_1} + \frac{1}{\phi_1}y_{t+2} - \frac{\epsilon_{t+2}}{\phi_1}\right) - \frac{\epsilon_{t+1}}{\phi_1} \\
&= -\frac{\phi_0}{\phi_1}\left(1 + \frac{1}{\phi_1}\right) + \frac{1}{\phi_1^2}y_{t+2} - \frac{\epsilon_{t+1}}{\phi_1} - \frac{\epsilon_{t+2}}{\phi_1^2} \\
&= -\frac{\phi_0}{\phi_1}\left(1 + \frac{1}{\phi_1} + \frac{1}{\phi_1^2}\right) + \frac{1}{\phi_1^3}y_{t+3} - \frac{\epsilon_{t+1}}{\phi_1} - \frac{\epsilon_{t+2}}{\phi_1^2} - \frac{\epsilon_{t+3}}{\phi_1^3} \\
&= \ldots \\
&= -\frac{\phi_0}{\phi_1}\left(1 + \frac{1}{\phi_1} + \frac{1}{\phi_1^2} + \cdots + \frac{1}{\phi_1^M}\right) + \frac{y_{t+M+1}}{\phi_1^{M+1}} - \frac{\epsilon_{t+1}}{\phi_1} - \frac{\epsilon_{t+2}}{\phi_1^2} - \frac{\epsilon_{t+3}}{\phi_1^3} - \cdots - \frac{\epsilon_{t+M+1}}{\phi_1^{M+1}} \qquad (8)
\end{aligned}$$

For a stationary solution, letting $M \to \infty$ and using $|\phi_1| > 1$ (so that $|1/\phi_1| < 1$), we get

$$y_t = \frac{\phi_0}{1-\phi_1} - \sum_{j=1}^{\infty} \frac{\epsilon_{t+j}}{\phi_1^j} \qquad (9)$$

It can be checked that (9) is also stationary (note now that $|\phi_1| > 1$) and satisfies the AR(1) equation $y_t = \phi_0 + \phi_1 y_{t-1} + \epsilon_t$. This AR(1) process is called **non-causal** because y_t depends on future values $\epsilon_{t+1}, \epsilon_{t+2}, \ldots$. Here ϵ_t is **not** independent of the past y-values $y_{t-1}, y_{t-2}, \ldots$.

If $|\phi_1| = 1$ (i.e., if $\phi_1 = 1$ or $\phi_1 = -1$), then neither (6) nor (8) converge as $M \to \infty$. In fact, in this case ($|\phi_1| = 1$), there cannot be a stationary solution to the AR(1) equation.

3.1 Practical Implications

In practice, while fitting the AR(1) model, we use the `AutoReg` function (from the `statsmodels` library) which works with the likelihood:

$$\prod_{t=2}^{n} \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{(y_t - \phi_0 - \phi_1 y_{t-1})^2}{2\sigma^2}\right) \qquad (10)$$

This likelihood fixes the value of y_1 and assumes that each ϵ_t is independent of the past values $y_{t-1}, y_{t-2}, \ldots$. In other words, the model considered is:

$$y_1 = \text{fixed at observed value} \quad \text{and} \quad y_t = \phi_0 + \phi_1 y_{t-1} + \epsilon_t \qquad (11)$$

where $\epsilon_t \overset{\text{i.i.d}}{\sim} N(0, \sigma^2)$ and independent of $y_{t-1}, \ldots, y_1$ for each $t = 2, \ldots, n$. The parameters ϕ_0, ϕ_1, σ are estimated in the usual way (as is done in linear regression). For predictions, one works with the fitted model where the parameters are replaced by their estimates $\hat{\phi}_0, \hat{\phi}_1$ (and also $\hat{\sigma}$).

Based on the value of $\hat{\phi}_1$, the following things can happen:

1. **Case One:** $|\hat{\phi}_1| > 1$. This happens often when we fit the AR(1) model directly to economic data (such as GDP or GNP). Here the fitted model (11) (with $\phi_0 = \hat{\phi}_0$ and $\phi_1 = \hat{\phi}_1$) is very different from the non-causal stationary AR(1) model given by (9). Note again that for (9), ϵ_t becomes dependent on the past values $y_{t-1}, y_{t-2}, \ldots$. The fitted model (11) is quite far from being stationary and future predictions will often exhibit explosive behavior when the prediction horizon increases.

2. **Case Two:** $|\hat{\phi}_1| < 1$. This also happens very often. For economic data, this often happens after some preprocessing (e.g., by taking logarithms of the raw data and then differences once or twice). In this case, the fitted model (11) is also non-stationary and distinct from the causal stationary model (7). However, the discrepancy is minimal and they behave similarly in many ways:

   a) The likelihood for (7) is

   $$\frac{\sqrt{1-\phi_1^2}}{\sqrt{2\pi}\sigma} \exp\left(-\frac{1-\phi_1^2}{2\sigma^2}\left(y_1 - \frac{\phi_0}{1-\phi_1}\right)^2\right) \prod_{t=2}^{n} \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{(y_t - \phi_0 - \phi_1 y_{t-1})^2}{2\sigma^2}\right) \tag{12}$$

   The only difference between this likelihood and the likelihood (10) is the presence of the term involving y_1. When n is large, this single term (which only involves the first data point) does not affect the overall likelihood significantly, so both likelihood maximizations lead to similar answers.

   b) Recursing the difference equation in (11) from $t, \ldots, 2$, we obtain (basically taking $M = t - 2$ in (6))

   $$y_t = \phi_0 \sum_{j=0}^{t-2} \phi_1^j + \phi_1^{t-1} y_1 + \sum_{j=0}^{t-2} \phi_1^j \epsilon_{t-j} \qquad \text{for every } t = 2, \ldots, n.$$

   Because $|\phi_1| < 1$ (so that $|\phi_1|^j$ decreases rapidly), we can expect the above to be close to (7) as long as t is not very small.

   c) Future predictions for the models (11) and (7) work in identical fashion (for fixed identical parameter values). These predictions only use independence of ϵ_t and past y-values $y_{t-1}, y_{t-2}, \ldots$ which is true in both (11) and (7).

   Because of these reasons, when $|\hat{\phi}_1| < 1$, even though `AutoReg` is fitting (11), it is common to pretend as if we are working with the causal stationary AR(1) model (7).

3. **Case Three:** $|\hat{\phi}_1| = 1$. This is either $\hat{\phi}_1 = 1$ or $\hat{\phi}_1 = -1$. These models are all non-stationary. The case $\hat{\phi}_1 = -1$ almost never arises in practice (unless the data has a strange wild oscillatory pattern from each time point to the next). The case $\hat{\phi}_1 = 1$ is much more applicable. `AutoReg` may not give $\hat{\phi}_1 = 1$ exactly but it might give $\hat{\phi}_1$ that is close to 1. Note that when $\phi_1 = 1$, the AR(1) model equation can be rewritten as $y_t - y_{t-1} = \phi_0 + \epsilon_t$; this suggests preprocessing the data by taking successive differences $y_t - y_{t-1}$ and trying to fit models to this differenced data. AR(1) predictions with $\hat{\phi}_1 = 1$ grow linearly which may be well-suited for many datasets.

Thus from the practical perspective where we only consider causal models, stationary only corresponds to $|\phi_1| < 1$. (When $|\phi_1| > 1$, mathematically speaking, there is a stationary AR(1) model but this is non-causal and does not correspond to our likelihoods and fitting technique).

References

[1] Robert H. Shumway and David S. Stoffer. *Time Series Analysis and Its Applications: With R Examples*. Springer.